ARCHANA J

Biomedical Engineer | Fresher | Graduating 2026

CONTACT

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EDUCATION

B.E. Biomedical Engineering
Kongunadu College of Engineering & Technology
2022 – 2026 | CGPA: 8.62

HSC
St. Antony's Matric. Hr. Sec. School
2022 | 75.50

SSLC
St. Antony's Matric. Hr. Sec. School
2020 | 83.80

TECHNICAL SKILLS

Programming
Python | Java | MS Excel

Hardware & Embedded
IoT platforms,
Embedded C
Microcontrollers

Biomedical Domain
Human Anatomy & Physiology,
Biomedical instrumentation,
Clinical equipment maintenance

Regulatory Awareness
Medical coding (VACBM01),
Basic ISO 13485 & MDR knowledge

CERTIFICATIONS

- Diagnostic Imaging — Yale University (Coursera)
- NPTEL — Enclosure Design of Electronics Equipment (71)

SOFT SKILLS

- Communication & Team Collaboration
- Adaptability & Time Management

LANGUAGES

Tamil (Native) | English (Professional)

PROFILE SUMMARY

Biomedical Engineering undergraduate with hands-on experience in IoT-based patient monitoring, embedded systems, and clinical equipment handling. Interned at Saveetha Medical College and Codebind Technologies. Seeking a fresher role in biomedical engineering, an application specialist, or clinical engineering to develop safe, efficient, and innovative healthcare technologies.

ACADEMIC PROJECTS

BMI Smart Wearable Knee Brace with Exoskeletal Assistance

Tech: ESP32, IoT, Pressure Sensors

- Designed an adaptive exoskeleton knee brace monitoring body load and knee movement in real time using embedded sensors.
- Implemented IoT cloud data transmission with automated anomaly alerts to reduce joint stress and improve mobility.

Pediatric Remote Monitoring Control Using IoT

Tech: ESP32, IoT Server, Temperature & Pulse Sensors

- Built an ESP32-based system tracking pediatric temperature and pulse remotely with automated threshold alerts.
- Transmitted patient vitals to an IoT server, enabling remote access for healthcare personnel.

Non-Invasive Blood Group Analyzer Using Image Processing

Tech: Python, OpenCV, Image Processing

- Developed a needle-free blood group detection system using image processing, eliminating conventional agglutination testing.
- Reduced patient discomfort by replacing needle-based testing with a computer-vision analysis pipeline.

INTERNSHIPS & TRAINING

Hospital Training Intern- Saveetha Medical College & Hospital, Chennai

- Supported in maintenance and calibration of clinical equipment including ECG machines, ventilators, and patient monitors.
- Gained exposure to hospital biomedical department workflows, equipment lifecycle management, and safety protocols.

Embedded Systems Intern- Codebind Technologies, Coimbatore

- Worked on embedded C/Python programming for hardware prototyping with microcontrollers and sensor integration.
- Gained hands-on experience in circuit design and systematic debugging of embedded systems.

Additional Training: VACBM04 Biomedical Hardware Training | VACBM01 Medical Coding Course

PAPER PRESENTATIONS

- "Brain Implants" — SNS College of Technology, Coimbatore
- "Pediatric Remote Monitoring Control Using IoT" — National Engineering College, Kovilpatti

AWARD

First Prize — Circuit Debugging
P.A. College of Engineering and Technology, Pollachi.